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Prevention and characterization of thin film defects induced by contaminant aggregates in initiated chemical vapor deposition

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ABSTRACT

As initiated Chemical Vapor Deposition (iCVD) finds increasing application in precision industries like electronics and optics, defect prevention will become critical. While studies of non-ideal morphology exist in the iCVD literature, no studies investigate the role of defects. To address this knowledge gap, we show that the buildup of short-chain polymers or oligomers during normal operation of an iCVD reactor can lead to defects that compromise film integrity. We used atomic force microscopy to show that oligomer aggregates selectively prevented film growth, causing these hole-like defects. X-ray diffraction and optical microscopy demonstrated the crystallinity of the aggregates, pointing to a flat-on lamellar or mono-lamellar structure. To understand the origin of the aggregates, spectroscopic ellipsometry showed that samples exposed to the reactor consistently accrued low-volatility contaminants. X-ray photoelectron spectroscopy revealed material derived from polymerization in the contamination, while scanning electron microscopy showed the presence of defect-causing aggregates. We directly linked oligomeric/polymeric contamination with defect formation by showing an increased defect rate when a contaminant polymer was heated alongside the sample. Most importantly, we showed that starting a deposition at a high sample temperature (e.g., 50 °C) before reducing it to the desired setpoint (e.g., 9 °C) unilaterally prevented defects, providing a simple method to prevent defects with minimal impact on operations.

I. INTRODUCTION

Over the last two decades, initiated Chemical Vapor Deposition (iCVD) has emerged as a promising polymer thin film synthesis technique. Because iCVD is a single-step process that produces low roughness, conformal, high-purity films,[1] it is being leveraged to produce better antifouling surfaces,[2] porous polymer films,[3] passivating coatings,[4] and adhesives[5] (for a comprehensive review, see Gleason[1]). In particular, however, the high quality of iCVD films has inspired numerous potential commercial applications in industries like optical[6] and electronic materials,[7,8] where uniform films with low roughness are required.[9,10] In these industries, defects can reduce product quality, prevent miniaturization, and decrease manufacturing yield (directly reducing revenue). As a result, for well-established manufacturing techniques like inorganic CVD and PVD, a robust body of literature exists to address defect prevention.[11] Despite its emerging application in precision devices, however, the polymer CVD literature has done little to discuss defect formation and prevention.

Taking iCVD as an example, while the literature contains investigations of roughness and island growth caused by low surface energy materials[12] and substrates,[13] discussion of defects has historically been limited to the observation that iCVD is unsusceptible to the types of surface tension-induced defects that affect techniques like spin coating.[14] The ability to create high-quality, defect-free films is instrumental for precision applications of iCVD, such as soft electronics, where iCVD has been successfully incorporated into devices.[15–17] As iCVD continues to mature as a technology and a wider range of deposition conditions and monomer chemistries are implemented, the importance of filling the current gap in our

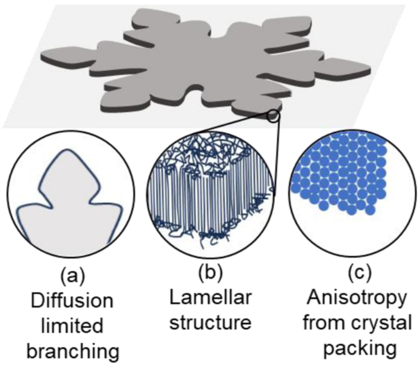

FIG. 1. Schematic showing a mono-lamellar polymer crystallite on a surface, highlighting (a) branching morphology driven by diffusion-limited aggregation, (b) flat-on mono-lamellar structure of ultrathin crystalline polymer films, and (c) horizontal cross-section of (b), showing crystalline packing of chains. Anisotropy and symmetry of the aggregate arise because new chains add to the rough tip more easily than they add to the flat prism facet. This phenomenon is also referred to as the surface tension anisotropy of the prism facet (see Sec. S2 of the supplementary material).

understanding of defect prevention will only grow. In the context of this paper, the term "defect" describes the optically observable structure caused by the contaminant in the polymer film, and the term "aggregate" describes the optically invisible contaminant that caused the defect to form.

Here, we demonstrate the existence of a previously unexplored defect type arising from crystalline aggregates with an in-plane fractal shape that is on the micrometer to millimeter scale. Our data indicate the aggregates are made up of oligomers that accumulate in the reactor during regular depositions. The morphology of such defects resembles surface-stabilized polymer crystals, which are known to form readily on solid surfaces and exhibit fractal structures (Fig. 1).[18–23] The morphology of the fractal structures is dictated by diffusion-limited crystallization [Fig. 1(a)] and the anisotropic surface tension of the crystal's prism facet [Fig. 1(c)].[24,25] Subtle differences in crystal anisotropy and supercooling can give rise to a variety of distinct in-plane fractal morphologies.[26,27] In particular, the complex in-plane fractal morphologies are understood via 2D diffusion-limited crystallization theories,[18] while the out-of-plane morphology is governed by the semicrystalline lamellar structure characteristic of polymer crystals [Fig. 1(b)].[28] In ultrathin crystalline polymer films (<100 nm), the polymer lamellae lay flat on the substrate surface.[18] Mono-lamellar and multi-lamellar polymer crystals can be differentiated using scanning electron microscopy (SEM) and atomic force microscopy (AFM)[21] since multi-lamellar crystals exhibit stepwise thickness changes in integer multiples of the lamellar thickness.[18]

So far, detailed reports of crystallinity in iCVD are limited to the crystallization of atactic poly(1H,1H,2H,2H-perfluorodecyl acrylate) (pPFDA), which crystallizes due to the bulkiness of its large perfluorinated side-chain.[29] In this case, the close-packing of the bulky side chains is the primary driver of crystallization, and the polymer backbone (including its stereochemistry) has little impact on the crystal structure.[30,31] While the semicrystalline nature of some pPFDA films synthesized in iCVD appears to affect the surface morphology of growing films, this phenomenon is distinct from the post-synthesis crystallization of polymeric material discussed herein.[32] Besides pPFDA,[29] reports of crystallinity in iCVD are exceedingly rare.[33] Indeed, NMR studies have confirmed the expected atactic[34] nature of iCVD polymers,[35] which are considered less prone to crystallization than their tactic counterparts.[34] While some of the defects we observed are associated with fluorinated polymers, others emerge in deposition environments that have never been exposed to fluorinated species, suggesting that such defects arise from atactic oligomer/polymer aggregates and crystallize by a different mechanism than atactic pPFDA.

Albeit uncommon, atactic polymers without bulky side chains have also been reported to crystalize—generally when factors are at play that allow locally tactic regions to find each other more easily. Furthermore, factors that enable greater chain extension can help overcome the energy barrier caused by chain entanglement,[36] as in the case of atactic poly(methyl methacrylate) crystallization in the presence of chain stretching agents.[37] Crystallization of poly(acrylonitrile) (PAN) and poly(vinyl chloride) (PVC) are both famous cases of atactic polymers that maintain relatively high crystallinity via enhanced interaction and packing among tactic segments that arise randomly within an atactic chain.[38,39] Crystallization of atactic oligomeric polystyrene has been observed by Chai *et al.* and is justified by considering that Bernoulli statistics would generate a sufficiently high concentration of stereoregular material to form crystallites.[23] In these three examples, chain stretching agents, conformational flexibility, and a low degree of polymerization play similar roles in enabling short sections of tactic polymer generated at random to aggregate and crystallize. Notably, free radial polymerization in iCVD is known to lead to a high-dispersity[40] and often low molecular weight polymer.[41] This type of polymerization would be expected to give rise to an oligomer presence, which resembles the conditions used by Chai *et al.* and may explain the defects observed herein.

In this study, optical microscopy and AFM microscopy were used to elucidate the process of defect formation from the oligomeric material accumulated in the reactor over the course of regular depositions. XRD was performed on heavily defect-laden samples to better understand the crystal structure. To elucidate the origins and nature of reactor contamination, spectroscopic ellipsometry clearly showed the presence of low volatility contaminants, while XPS and SEM of the contamination showed material derived from polymerization and aggregate formation, respectively. By heating a pPFDA sample alongside a cooled, clean wafer, we intentionally induced defect formation, illustrating that iCVD thin films can be a source of oligomeric contaminants. Finally, we show that defect-free films can be produced (even under conditions that would normally cause defects) when the deposition is begun at a high stage temperature to form a defect-free conditioning film, and the stage temperature is subsequently dropped to the desired deposition setpoint. For consistency, we have chosen to use poly(acrylic acid) (pAA) to resolve defects unless otherwise specified (such as samples in Fig. 2, which illustrate the presence of defects in a range of polymer types). We suspect that defects have been avoided historically by using relatively high stage temperature deposition conditions, explaining why defect formation has not been a problem for precision applications of iCVD to date.[1] As we show, defects can be seen clearly with optical

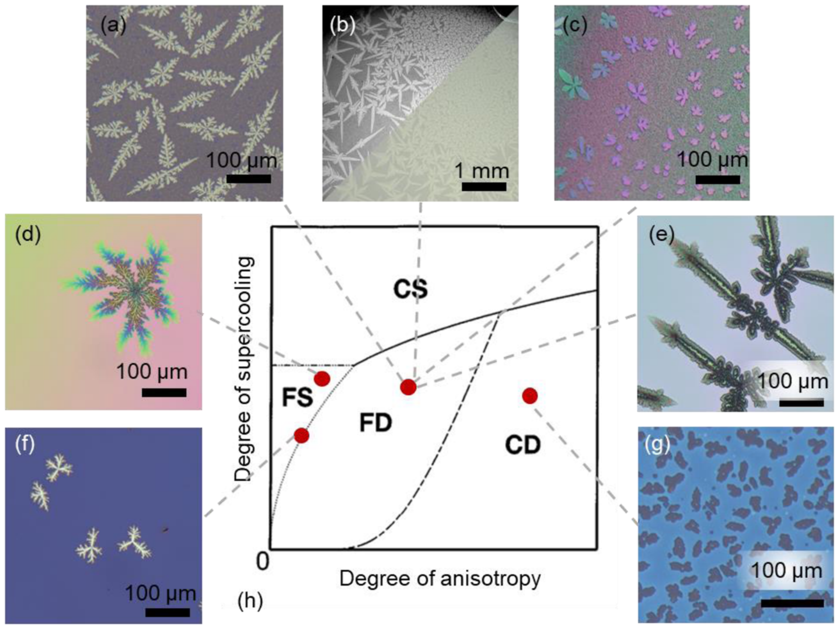

FIG. 2. (a) Micrograph of a p(AA-co-DEGDVE) film exhibiting a defect with the fractal dendrite morphology. (b) Micrograph of a pPFDA film exhibiting large fractal dendrite defects, indicating that defect formation occurs even with hydrophobic monomers like PFDA. The image was modified to clearly show defects (upper left), with the original image also provided (lower right). (c) Micrograph of defects formed on a pAA base-layer, resolved by a second layer of pAA film, showing that defects can form on a range of substrates, including silicon thermal oxide and polymer. (d) Optical micrograph of a p(4VP-*co*-DVB) film exhibiting a defect with the fractal seaweed morphology. (e) Micrograph of a p4VP film exhibiting fractal dendrite defects, showcasing the diversity of morphologies observed. (f) Micrograph of a pDVB film exhibiting defects with a transitional morphology between fractal dendrites and seaweeds. (g) Micrograph of a pAA film exhibiting a compact dendrite morphology (also in Fig. 7). (h) An example of a phase diagram taken with permission from Brenner *et al.*[27] The regions of the phase diagram describe relationships between morphology, labeled as compact seaweed (CS), compact dendrite (CD), fractal dendrite (FD), and fractal seaweed (FS), with the precise relationships depending on the system under study. Labels categorizing micrographs into these morphologies are an aid to the eye and do not correspond to measured or calculated anisotropy and supercooling. The degree of anisotropy describes the anisotropy of surface tension on the prism facet of a growing crystal (e.g., a higher degree of anisotropy indicates a higher surface tension on the prism edges); the degree of supercooling describes the driving force for crystallization (e.g., a higher degree of supercooling represents a greater driving force to crystallize).

microscopy because the film thickness variation they cause manifests vividly as coloration. However, optical microscopy has been rare in the iCVD literature compared with characterization by techniques like ellipsometry, SEM, and FTIR, which do not show defects clearly. By addressing the overlooked aspect of defect formation in iCVD, our research aims to contribute fundamental understanding and provide practical prevention measures to current and future industries that may rely on iCVD.

II. MATERIALS AND METHODS

A. Materials

4-vinylpyridine (4VP) (95%, from Sigma Aldrich), divinyl benzene (DVB) (80%, from Sigma Aldrich), di(ethylene glycol) divinyl ether (DEGDVE) (99%, from Aldrich), 1*H*,1*H*,2*H*,2*H*-perfluorodecyl acrylate (PFDA) (97%, from Sigma Aldrich), and acrylic acid (AA) (99%, from Sigma Aldrich) were used as monomers during depositions. Di-*tert*-butyl peroxide (TBPO) (98%, from Aldrich) was used as an initiator. All monomers and the initiator were used without further purification. All depositions were performed on polished p-type/boron-doped silicon wafers (from University Wafer). Argon was obtained from Airgas and was used as a patch gas to maintain a consistent overall gas flow rate in some depositions. Thermal grease (Céramique™ 2, a tri-linear ceramic thermal compound from Arctic Silver) was used to ensure thermal contact between the 1″x1″ thermoelectric cooling device (TEC) (from TE Technology' Inc.) and the silicon wafer.

B. Film synthesis

Polymer thin film synthesis occurred in a custom-built reactor. Samples were cut from silicon wafers, setting aside a reference sample in each case as a control for characterizations. Once samples were loaded into the reactor, it was closed off to the atmosphere and evacuated to base pressure (2–3 mTorr). During this time, an Accel 500LT recirculating chiller (Fisher Scientific) was used to cool the stage to its operating temperature (10–50 °C). For depositions requiring a sample temperature of −5 to 10 °C, the chiller was set to 15 °C and a TEC was used. After pumping down for one hour, the leak rate was recorded in sccm (standard cubic centimeters per minute). In all depositions, the leak rate was less than 0.05 sccm. At this stage, the flow rate of each monomer and the initiator were calibrated by closing the throttle valve and measuring the linear increase in pressure. In all depositions, TBPO was used as a thermal initiator, the flow rate of which was controlled using an MKS mass flow controller type 1159B. Monomer feed into the reactor was controlled using a needle valve.

After pump down, all flow rates and pressure were brought up to their desired setpoints and allowed to equilibrate for 5 min. Pressure in the reactor was monitored using an MKS Baratron capacitance manometer model 626C and was controlled using an MKS throttle valve (model no. 253B) at the outlet of the reactor. Once equilibrated, the filament array was turned on and brought up to 215 °C. The filament was resistively heated using a BK Precision DC power supply, and its temperature was monitored with a type k thermocouple during the deposition. The thickness of films was monitored during deposition using a 633 nm *in situ* laser interferometer with a JDSU He/Ne gas laser (model number 1508P-1). Once the desired thickness was reached, the deposition was stopped by turning off the filament array and evacuating the reactor. Refer to Table S1 for supplementary information on deposition conditions. All pPFDA depositions were performed using the same deposition conditions, and pAA was used as a standard polymer in this study to resolve defects for consistency between experiments.

C. Native aggregate synthesis

Native contaminant aggregates reported in Sec. III C were prepared according to the same procedures as a normal deposition (Sec. II B), including leak testing, monomer and initiator flow rate testing, and pressure equilibration. The only differences from normal deposition were that the filament array was not turned on (no film was deposited), and because the goal of these experiments was to observe contamination, an additional 2 h were given for cooled samples to collect contaminants (see synthesis conditions in supplementary information).

D. Polymer distillation

Polymer distillation experiments required two consecutive depositions on the same day. First, five roughly 2 cm × 4 cm rectangles were cut from a single silicon wafer, three for the control deposition and two for the polymer distillation itself. For the polymer distillation, an additional pPFDA coated wafer from a previous pPFDA deposition (conditions reported in Table S1) of the same size was also used. For both the control deposition and polymer distillation, one of the wafers is set aside as a blank. For the control deposition, two TEC plates are placed adjacent to each other in the reactor, and both remaining blank samples are placed on the TEC plates, as shown in Fig. 6(a). TEC temperatures are monitored with type k thermocouples affixed to samples with Kapton tape and thermal grease and are controlled manually using a high voltage power supply. The reactor is pumped to base pressure, flow rates are calibrated, and the reactor is brought up to target conditions as normal. Before turning on the filament array and starting the deposition, the voltages are set on the TEC plates so that one heats to 45 °C (sink wafer), while the other simultaneously cools to 0 °C (source wafer). This heating step drives off any aggregates from the sink that may have formed during pump-down. The voltages are then switched so that the temperature of the sink sample drops to 0 °C and the source sample rises to 45 °C, stabilizing after roughly 30 s. The sink sample is then allowed to sit for 30 min at target pressure with gases flowing and collecting aggregates. After the collection period, the filament array is turned on, and deposition is performed to resolve any aggregates that formed on the sink sample.

After the control deposition is performed and samples are removed from the reactor, a blank wafer (sink) is secured to the cold TEC using thermal grease, and the pPFDA coated wafer (source) is secured to the hot TEC using thermal grease. The procedures outlined earlier for the control deposition are then repeated so that the pPFDA sample is heated and the blank wafer is cooled. This enables the transfer of pPFDA from the hot (source) wafer to the cold (sink) wafer. Statistical significance between the sample and controls was demonstrated with a paired t-test with a one-tailed t-distribution.

E. Conditioning film method

To ensure rapid sample temperature change, a TEC plate is used. The TEC plate and sample are placed in the reactor, and good heat transfer is maintained with thermal grease. The normal deposition procedure is followed with the chiller set to a temperature of 20 °C and the TEC plate set to a temperature of 50 °C. Once the film thickness reaches 100 nm, the temperature is lowered to 9 °C by switching the voltage on the power supply and manually controlling it until the setpoint is reached. The rest of the deposition is performed as normal.

F. Characterization and data analysis

A VHX-970F digital microscope from Keyence collected optical micrographs of films with appropriate magnification to capture the defect scale. The ImageJ particle analysis program was used to quantify defect surface coverage for polymer distilled samples and controls synthesized as described in Sec. II D. To show statistical significance between defect populations generated in Sec. II D, a one-tailed paired t-test was performed on surface coverage values from particle analysis in ImageJ.

An Asylum Research MFP-3D-BIO atomic force microscope (AFM) was used in AC tapping mode to analyze the morphology of surfaces containing crystalline structures. A soft tapping mode AFM probe (PPP-NCSTR, NANOSENSORS) was used to scan a 30 × 30 μm region at 0.5 Hz. A 10 × 10 μm region was used for better resolution of island growth in the scan of fully developed defects.

A Bruker D8 Advance ECO powder diffractometer collected powder x-ray diffraction (XRD) scans of samples with high defect loading. Scans were performed over an angle range from 2° to 16° from a 1 kW Cu–Kα x-ray source.

A JA Woolam alpha-SE spectroscopic ellipsometer collected spectra at 65°, 70°, and 75° angles of incidence on aggregate samples and their respective blanks. Data were fitted using Complete EASE 6 software version 6.51 with a 3-layer model of a Cauchy film resting on a silicon thermal oxide on a semi-infinite silicon metal substrate. For contaminant samples synthesized according to Sec. II C, statistical significance between reference samples and contaminant samples was demonstrated using a one-tailed paired t-test.

A ScientaOmicron ESCA 2SR XPS collected x-ray photoelectron spectroscopy (XPS) survey scans from aggregate samples prepared using the procedures in Sec. II C. The x-ray source pass energy was 150 eV, and the filament voltage was 10 kV. Data were collected with a step size of 1 eV. Data analysis and peak integration were performed using the CasaXPS software.

A Zeiss Gemini 500 Scanning Electron Microscope (SEM) collected SEM micrographs of aggregate samples.

A Nicolet iS50 FTIR spectrophotometer with a variable angle grazing attenuated total reflectance (VariGATR) attachment set at 60° and a liquid nitrogen cooled mercury-cadmium-telluride (MCT) detector measured the infrared spectra of aggregate samples prepared using the procedures in Sec. II C. Spectra were collected from 700 to 4000 cm^{-1} with a resolution of 4 cm^{-1}. Omnic software was used for data analysis, a and built-in atmospheric correction was used to eliminate spurious peaks where necessary. To avoid bias, a baseline correction was not performed (FTIR spectra were inconclusive and may be found in Fig. S2 of the supplementary material).

III. RESULTS AND DISCUSSION

A. Defect characterization by optical and atomic force microscopy

A selection of optical micrographs depicting the thin film defects and their shapes under study are displayed in Fig. 2. Adopting the language used in the field of diffusion-limited aggregation, aggregates that show anisotropy are called "dendrites," while those that do not are called "seaweeds." Similarly, aggregates that form branching structures are called "fractal," while those that form islands with no branching are "compact."[26,27] For background on diffusion limited crystallization theory, we direct the reader to Chapter 6 of Saito.[24] An excellent review by Liu and Chen describes the application of this theory to 2D polymer crystallization.[18]

Defects shown in Figs. 2(a)–2(c) and 2(e) are good examples of fractal dendrites [FD in the phase diagram in Fig. 2(h)]. The fractal dendrite morphology indicates that contaminant aggregates formed at a moderate degree of anisotropy (caused by moderate surface tension anisotropy on the crystal's prism facet) and low supercooling (i.e., the driving force for crystallization).[27] Though these samples are all classified as fractal dendrites, their diversity indicates sensitivity to growth conditions or variability in aggregate composition. Figure 2(d) shows an example of a fractal seaweed (FS in the phase diagram), expected to form under conditions of low anisotropy.[26,27] Compact seaweeds (CS), which form under a high degree of supercooling, were not observed under our reactor conditions. Figure 2(g) shows an example of a compact dendrite (CD in the phase diagram) with no branching. Compact dendrites form with a high degree of anisotropy. Figure 2(f) exhibits a transitional morphology, indicating it was formed at conditions close to the coexistence between dendrites and seaweeds. Figure 2(b) shows the largest single defects we have observed (which were resolved using a pPFDA film), showing numerous millimeter-scale defects (the upper left half of the micrograph was modified to highlight defects, with the original image provided on the lower right).

It is important to note that the labels categorizing micrographs in Figs. 2(a)–2(g) are purely based on morphology and do not reflect estimates of supersaturation and anisotropy. Though the degree of supercooling and degree of anisotropy are well-defined quantities in a theoretical context and are, therefore, instrumental in generating a phase diagram like that shown in Fig. 2(h),[27] the conventional iCVD setup used to synthesize samples in Fig. 2 is not capable of controlling aggregation conditions enough to estimate these parameters. The degree of supercooling is a dimensionless metric of the driving force for aggregation, and the degree of anisotropy is a dimensionless metric of how strongly surface tension is affected by the crystal facets[24,27] (explicit theory is presented in the supplementary material and is discussed in more detail in Saito[24] and Brener *et al.*[27]).

Beyond morphology, optical micrographs can provide some important insights into the nature of defects. Figures 2(a)–2(c) and 2(g) show samples synthesized in a reactor that was frequently exposed to fluorinated monomers, whereas Figs. 2(d)–2(f) show defects that were deposited in a reactor only exposed to chemistries that had been used without incident in other reactors and were not fluorinated. Observing defects in a fluorine-free reactor indicates that fluorinated material is not a prerequisite for defect formation, leaving built-up polymeric/oligomeric material from past depositions as a likely culprit. While all other micrographs show defects on a silicon wafer (with the top layer being an amorphous silicon thermal oxide), Fig. 2(c) shows defects on a (defect-free) pAA film (resolved by a second pAA deposition), demonstrating that defects can form on a range of surfaces.

The defect formation mechanism was illustrated using AFM scans of defects at subsequent stages of development (Fig. 3). Defect-causing aggregates formed after samples were loaded into the iCVD reactor and during the pump-down step while stage and line temperatures stabilized. The AFM scans of the aggregates that formed after the pump-down step are shown in Figs. 3(a)–3(c). After the reactor reached base pressure (1–3 mTorr), flow rates of the initiator and monomer were calibrated, the chamber pressure was set, and the deposition was started by turning on the filament array (see Sec. II B and Table S1 for methodological details and deposition conditions). After 5 min of depositing a pAA film, growth only occurred on the pristine surface, and depressions began to form where the aggregates were located [Figs. 3(d)–3(f)]. By the end of the deposition, the aggregate surface remained film-free but exhibited islands where polymer growth nucleated [Figs. 3(g)–3(i)]. These islands of polymer growth resembled those observed by Rumrill *et al.*,[13] where poly(hydroxyethyl methacrylate) islands were deposited on a poly(tetrafluoroethylene) film, implying a similar mechanism for island formation (i.e., driven by surface energy mismatch). Defects were visible in optical micrographs (Fig. 2) because even small variations in thickness or roughness cause significant color changes. The "rainbow" appearance of defects in Figs. 2(d) and 2(e) is also explained by this correspondence between color and thickness and the transformation of defects from holes to depressions in the film as islands merge. These results demonstrate clearly that hole formation is caused by the inhibition of deposition on the contaminant surface as opposed to the space-filling role played by immiscible small molecules in porous film formation from iCVD.[42–45]

B. XRD and anisotropy

To fully understand defect morphology, it is necessary to discuss the anisotropy intrinsic to our defects. Because the silicon substrate and polymer film are both amorphous, the anisotropy in fractal dendrites (Fig. 2) is attributed to the crystallinity of the defect-causing aggregate. As such, we performed powder XRD on a sample that exhibited prodigious defect formation (Fig. 4, top trace and inset). Because of the orientation of the crystal relative to the diffractometer, all d-spacings represent length scales normal to the surface. The observed d-spacing (3.75 nm) is consistent with the aggregate thickness observed in AFM traces [3.7–4 nm, Fig. 3(a)]. Assuming the aggregates are oligomeric/polymeric in nature, this d-spacing corresponds well to the reported lamellar spacing.[18] Furthermore, the presence of diffraction in this sample as well as the remarkably high defect loading indicate a potentially multi-lamellar structure, in contrast to other samples analyzed with SEM and AFM (Figs. 3 and 5). Given the similarity of diffraction

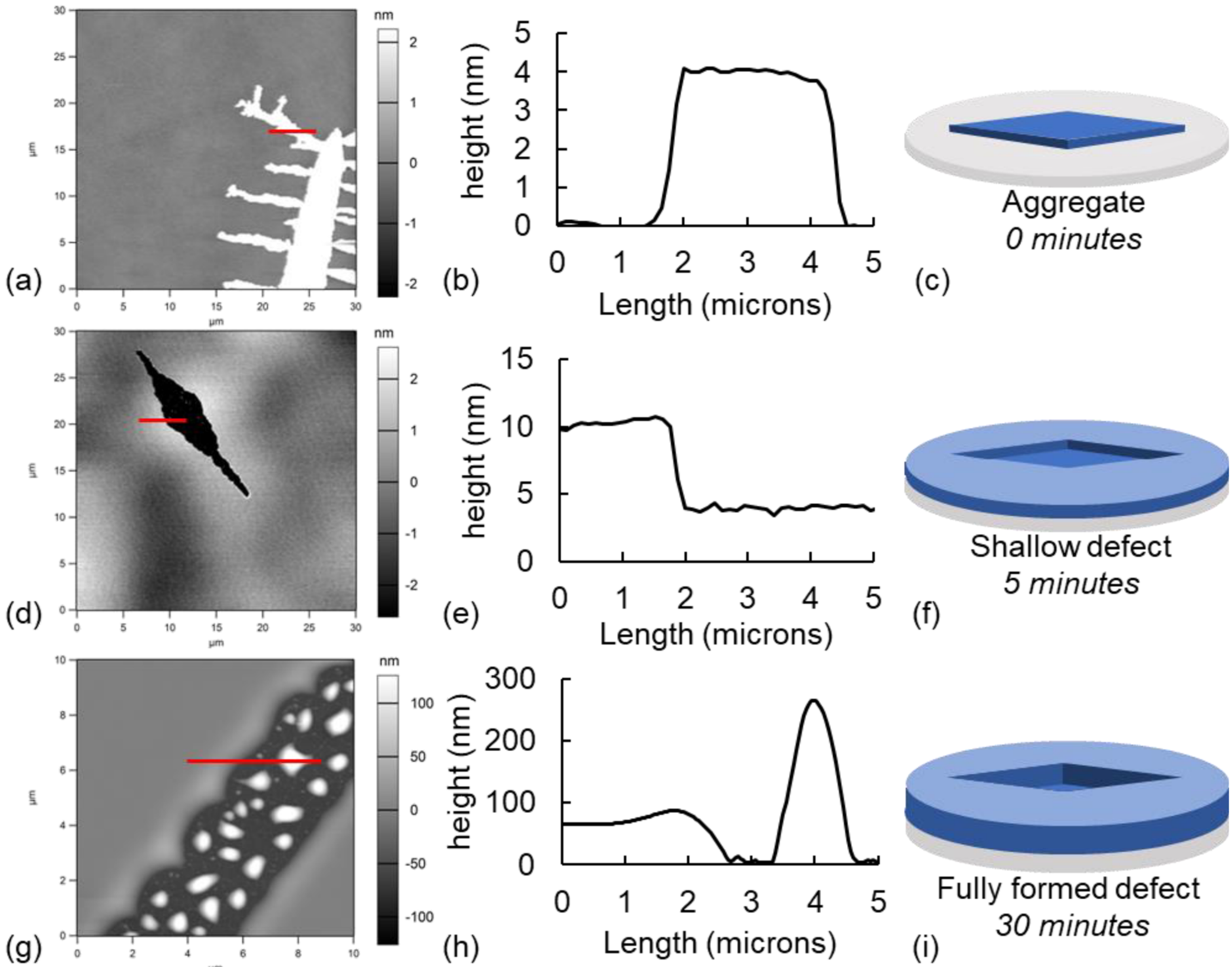

FIG. 3. (a) AFM scan of a contaminant aggregate, where the red bar indicates where the line scan in panel (b) was taken (all line scans run left to right). (b) Line scan of a contaminant aggregate shown in panel (a). (c) Cartoon showing the contaminant aggregate represented in panel (a). (d) AFM scan of a shallow defect in a pAA film (deposition conditions in Table S1), where the red bar indicates where the line scan in panel (e) was taken. (e) Line scan of the shallow defect shown in panel (d). Pristine film thickness was normalized to 10 nm based on spectroscopic ellipsometry measurements. (f) Cartoon showing the shallow defect represented in panel (d). (g) AFM scan of a fully developed defect in a pAA film (deposition conditions in Table S1), where the red bar indicates where the line scan in panel (h) was taken. (h) Line scan of the defect shown in panel (g) showing large islands of iCVD polymer on the defect surface. (i) Cartoon showing the fully developed defect represented in panel (g).

patterns, the crystallinity observed for iCVD poly(ethylene glycol dimethacrylate) and poly(isobornyl acrylate)[33] may also be due to lamellar structures.

Crystallinity in iCVD films has been observed most strongly for pPFDA, with the crystallization explained by the packing of the bulky side chains.[29,30] Furthermore, fluorine was found in some contaminant material (Sec. III C), although not all [Figs. 2(d)–2(f)]. Therefore, we believe there is enough reason to assess the possibility that pPFDA may be the cause of defects in Fig. 4. Based on the powder XRD data (Fig. 4, top and bottom traces), there is no correspondence between the crystalline pPFDA film and that of the contaminant. Powder XRD patterns distinct from pPFDA have also been reported for poly(ethylene glycol dimethacrylate) and poly(isobornyl acrylate) synthesized by iCVD,[33] though the origin of the crystallinity was not discussed.

C. Surface characterization of contaminant material

While optical and AFM micrographs show that defects were caused by contaminant aggregates, they provide few insights into the origins of contamination. To characterize the contaminant material present in the iCVD reactor, the contaminant was collected by placing three samples on the cooled stage for 3 h. The next day, ellipsometry was conducted on samples exposed to contaminants in

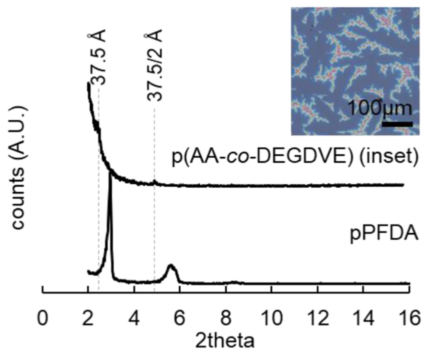

FIG. 4. Powder XRD of contaminated p(AA-*co*-DEGDVE) film (top trace) shown in inset, crystalline pPFDA (micrograph not shown). The sample in the inset was produced from a pAA deposition and shows crystallinity with a d-spacing of 37.5 Å. Note that the powder diffraction pattern for the sample shown in the inset does not match pPFDA.

the reactor and blank samples not exposed to the reactor. SEM and XPS were performed on samples after the exposure (Figs. S1–S3 and Tables S2 and S3 in the supplementary material).

Spectroscopic ellipsometry data were fitted to a model representing a Cauchy film on silicon thermal oxide. The three Cauchy-film thicknesses for samples not exposed to the reactor averaged 0.46 ± 0.01 nm (mean ± standard error). This thickness value represents a layer of ambient contamination (e.g., adsorbed water) or model error and should be interpreted as a baseline as opposed to an actual "film." Ellipsometry was also performed on the contamination-exposed samples (i.e., on wafers identical to the pre-exposure measurements). The average thickness was 0.87 ± 0.06 nm (mean ± standard error). The difference is statistically significant, with $p = 0.011$ from a 1-tailed paired t-test, showing that samples that spent time in the reactor picked up contaminant material. Because post-exposure samples were removed from the reactor a day before they were characterized, the contaminant material must be stable and non-volatile, thus likely macromolecular. The simplest source for such a contaminant is oligomeric material accumulated from past depositions.

SEM images of the post-exposure wafers revealed aggregate formation on two of them, with some aggregates appearing to be in the process of melting or sublimating [Figs. 5(a) and 5(b)]. XPS survey scans performed on the same samples showed fluorine in 2 of the 3 samples (Fig. S1 and Table S3 in the supplementary material). There was no notable correlation between samples with fluorine and observable aggregates in SEM. Observation of aggregates on samples never directly exposed to monomer vapor and showing stability under ambient conditions for hours to days eliminates the possibility that monomer crystals or small molecule crystals are involved in defect formation. Because synthesizing samples in a clean room setting was infeasible in this study, the obscuring presence of ambient contamination confounds a more detailed analysis of the XPS data. FTIR was also collected for these samples (Fig. S2 in the supplementary material) but was inconclusive.

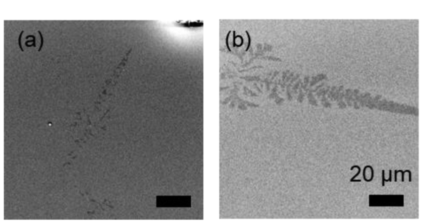

FIG. 5. SEM images depicting remnants of melting or sublimating contaminant aggregates prepared using the procedures described in Sec. II C. (a) Almost fully melted or sublimated aggregate. (b) Aggregate showing moderate melting or sublimation.

We note that most surface analysis techniques are not sufficiently sensitive to characterize nanogram quantities of material (outside a cleanroom environment). This includes techniques like SEM with electron dispersive spectroscopy and polarized light microscopy. Similarly, bulk chemical analyses like nuclear magnetic resonance spectroscopy and matrix assisted laser desorption/ionization time of flight mass spectrometry require milligram quantities of material, making such techniques infeasible. The reproducibility of these techniques is further jeopardized by non-defect-forming contamination (such as amorphous oligomers), contamination from the characterization technique itself (adventitious carbon), or damage to the aggregates by x rays or electron beams. Considering these pitfalls, we can confidently draw two conclusions based on the surface analyses. First, samples that spend time in the reactor consistently and repeatably accrue non-volatile contaminants capable of remaining on the surface for at least 1–2 days. Second, fluorinated chemistry is often present in contamination. Because fluorine is only introduced to the reactor as a constituent of monomers, its prolonged presence in the reactor means that some fraction of the contaminant material captured by ellipsometry is derived from polymerization and is most likely oligomeric or polymeric. In Sec. III D, an indirect method is discussed, which circumvents the limitations of surface analysis and enables a well-controlled study of defect formation.

D. pPFDA polymer distillation

While material(s) other than pPFDA can cause defects, as suggested by optical microscopy and XRD (Figs. 2 and 4), in this section, we demonstrate an example where defect formation was induced via exposure to pPFDA. The experimental setup used is shown in Fig. 6(a), and detailed methods can be found in Sec. II D. Briefly, a pPFDA-coated "source wafer" was heated alongside a clean and uncoated "sink wafer" for 30 min, after which a pAA deposition was performed on both wafers. To obtain a baseline of defect formation in the reactor, the experiment was also performed with an uncoated source wafer (on the same day). Select optical micrographs of the sink wafers for pPFDA and blank sources are provided in Figs. 6(b) and 6(c), respectively. This experiment was performed three times, showing a statistically significant increase in surface coverage with $p = 0.027$ from a one-tailed t-test (see Fig. S4 in the supplementary material for all micrographs). Table I shows the defect surface coverage for each distillation experiment and its control (quantified using particle analysis in ImageJ).

Having shown in this section that pPFDA can form defects and in Secs. III A and III B that defects also occur without pPFDA (Figs. 2

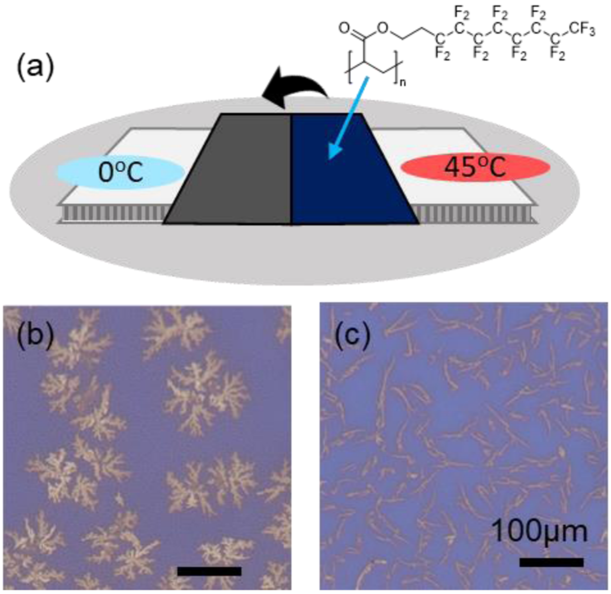

FIG. 6. (a) Experimental setup showing polymer distillation of pPFDA from a wafer placed on a hot TEC at 45 °C to the cold wafer placed on a TEC cooled to 0 °C. (b) Optical micrograph of defects caused by pPFDA distillation. (c) Optical micrograph of the control sample for panel (b), showing baseline reactor contaminant levels.

and 4), we have demonstrated that multiple chemical species are responsible for defect formation. Furthermore, because the pPFDA-derived defect morphology observed in this distillation experiment matches defects that arise spontaneously [Fig. 2(d)], this experiment supports the conclusion that defects are caused by materials derived from polymerization—specifically a build-up of oligomeric material. More *in situ* distillation experiments and ex-situ investigations of oligomer crystallization[23] are necessary to better understand what properties make a given oligomer inclined to form defect-causing crystalline aggregates. However, that is outside the scope of this study.

E. Defect prevention by the conditioning film method

High-quality film synthesis using iCVD will require a procedure to avoid defect formation. Since entropy of crystallization is generally negative, crystalline aggregates are expected to form more readily at lower temperatures. To confirm this, pAA depositions were performed at a range of stage temperatures and were characterized with optical microscopy (see Table S1 for deposition conditions

TABLE I. Defect coverage data for pPFDA distillation.

Run no.	Control defect coverage (%)	Sample defect coverage (%)
1	0.6	18.5
2	7.3	39.7
3	7.8	23.2

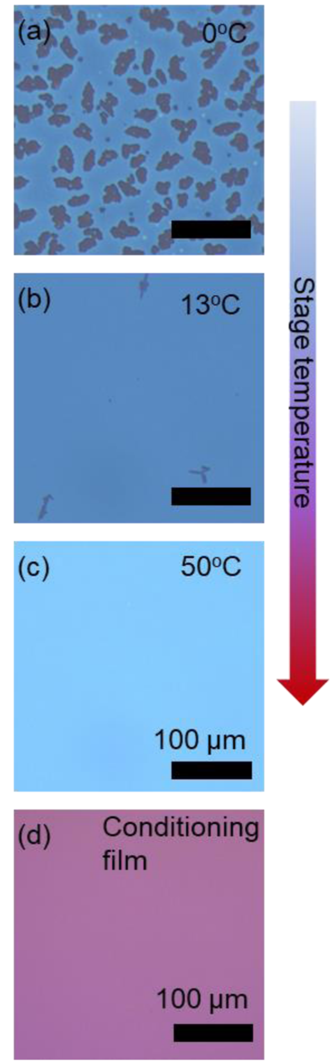

FIG. 7. (a) Optical micrograph showing defects resolved by pAA using a stage temperature of 0 °C. (b) Optical micrograph showing defects resolved by pAA using a stage temperature of 13 °C. Noticeably fewer defects are present than panel (a). (c) Optical micrograph showing a defect-free pAA film using a stage temperature of 50 °C. (d) Optical micrograph of a film made using the conditioning film method (see Sec. II E), showing the absence of defects.

and Sec. II B for methods). We found that, as the stage temperature increased from 0 to 13 °C, aggregate formation was suppressed [Figs. 7(a) and 7(b)]. At 50 °C, no defects were observed [Fig. 7(c)]. Because the stage temperature is often reserved as a key synthesis condition that regulates the kinetics of iCVD polymerization, performing all depositions at such a high stage temperature may not be feasible to prevent defects.

Because the focus of this section is defect prevention during normal reactor operation, the samples shown in Figs. 7(a)–7(c) were synthesized using normal deposition procedures, so their defect coverage necessarily depends on the unknown contaminant loading, which is subject to moderate variation over time. The trend shown in Figs. 7(a)–7(c) is a qualitative verification of the well-established thermodynamic principle that crystal formation is inhibited at elevated temperatures since entropy of crystallization is generally negative. A quantitative study of the dependence of defect formation on temperature requires good controls, which can be achieved using the distillation procedures discussed in Sec. III D but are nontrivial and outside the scope of this study.

We demonstrate a generalizable approach to prevent defect formation, which consists of starting a deposition at a high stage temperature (e.g., 50 °C) and, once the deposition starts (filament temperature reaches set-point), lowering the stage temperature without stopping the deposition (e.g., to 5 °C), to obtain defect-free films at conditions that might usually be prone to defects [Fig. 7(d)]. The thicknesses reported by AFM in Fig. 3, coupled with consistent defect coloration within optical micrographs in Fig. 2 (where light interference relates color and thickness), indicate consistent defect depth. This implies that all defects were formed at the start of a deposition, and defects do not form on actively growing films. Therefore, high stage temperatures only need to be used at the start of deposition and do not need to be imposed throughout. According to this logic, the defects studied here can always be prevented, provided the onset of a deposition utilizes a set of conditions that are not conducive to defect formation. We call this technique the conditioning film method since it relies on establishing an actively growing defect free film that conditions the surface against defects.

IV. CONCLUSIONS

We have introduced evidence for a new defect type that has the potential to impact low-stage temperature depositions (<25 °C) using the iCVD technique. AFM on samples obtained at various stages of defect development showed that contaminant aggregates form on the sample surface before deposition and cause holes in the final film by preventing film growth on the aggregate surface. Optical micrographs of defects match theoretical expectations for diffusion-limited aggregation.[26] Observation of anisotropic defects on multiple types of amorphous surfaces shows that defect-causing contaminant aggregates are made of crystalline material and do not originate from flaws in the substrate.

XRD was performed on a highly defect-laden p(AA-*co*-DEGDVE) film to better understand the underlying crystal structure. The resulting XRD patterns showed peaks corresponding to a d-spacing of 3.75 nm, which matched the 3.7–4 nm lamellar thickness measured with AFM. Spectroscopic ellipsometry demonstrated that samples exposed to the reactor for 3 h took up low volatility contaminants stable in air for at least 1–2 days. XPS survey scans detected fluorine in 2 of the 3 samples, indicating the presence of material derived from polymerization. SEM images confirmed the presence of aggregates on 2 of the 3 samples, notably showing melting or sublimating aggregates. By heating a source of polymer (pPFDA) alongside a cooled wafer and then performing a pAA deposition, we confirmed that polymeric material can cause defects. The simplest conclusion to draw from these data is that defects are caused by a build-up of oligomeric material from regular iCVD operations. This is supported further by the crystallization of atactic oligomeric material under similar circumstances[23] and our expectation that iCVD polymerization will generate significant quantities of oligomeric material.[40]

To prevent defects, we show that they occur less frequently at high sample temperatures, and if a deposition is started at a high sample temperature, it will be defect-free regardless of whether the temperature is subsequently dropped to a much lower value throughout the rest of the deposition. We expect this procedure to be especially valuable for applications of iCVD that require high-quality, defect-free films. It should be noted that this "conditioning" film method represents only one method by which defects can be prevented. The ability of a contaminant to crystallize on the surface likely depends on the surface chemistry as well as temperature, so depending on the nature of the contaminant oligomer material, certain substrate surface chemistries may be effective at lowering the sample temperature at which defects proliferate. A similar method has been used to prevent roughness in iCVD under supersaturated conditions.[3] Furthermore, if cleaning procedures can be developed that prevent the buildup of oligomeric material, this could also be a valid avenue to prevent defects.

In the future, it will be necessary to survey a range of potential contaminant materials and molecular weights to better understand the risk posed by this new defect type to the precision application of iCVD. We assume that some applications of iCVD will be unaffected by this defect type and would suggest future work to address the impact of such defects in the context of intended applications. Future experimental and theoretical work should be performed to elucidate the relationships between monomer adsorption and contaminant aggregation as well as determine why aggregates inhibit film growth. Finally, we hope that the methods outlined here might enable a lower cost or higher throughput study of diffusion-limited aggregation by providing a simple way to observe large numbers of polymer aggregates, which might be difficult with AFM methods.

SUPPLEMENTARY MATERIAL

The supplementary material contains supplementary data in the form of tables and figures and a short discussion of diffusion limited aggregation theory. The section covering supplementary tables and figures contains a table of deposition conditions used for film synthesis (Table S1), XPS survey scans for results reported in Sec. III C (Fig. S1), as well as a table of atomic mole ratios estimated from the XPS data (Table S2), FTIR spectra for results reported in Sec. III C (including blank and water spectra) (Fig. S2), thicknesses from ellipsometry for results reported in Sec. III C (Table S3), SEM micrographs for results reported in Sec. III C (Fig. S3), and optical micrographs of samples reported in Sec. III D (Fig. S4).

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Simon Shindler: Conceptualization (equal); Data curation (lead); Methodology (lead); Writing – original draft (lead); Writing – review & editing (equal). **Trevor Franklin**: Data curation (supporting). **Rong Yang**: Conceptualization (equal); Funding acquisition (lead); Supervision (lead); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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